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Task

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Abstract

Remote sensing and satellite image classification have become critical tools for environmental monitoring, urban planning, and agricultural management. This project presents a comprehensive end-to-end deep learning pipeline for classifying satellite images from the EuroSAT RGB dataset into ten distinct land-use and land-cover categories: AnnualCrop, Forest, HerbaceousVegetation, Highway, Industrial, Pasture, PermanentCrop, Residential, River, and SeaLake. The study is divided into two major parts. In Part A, two convolutional neural network architectures are designed and trained from scratch — a baseline model comprising three convolutional-pooling blocks and three fully connected layers, and a deeper model that doubles the convolutional depth and incorporates Batch Normalization and Dropout for regularization. Both models are compared across accuracy, loss, training time, and evaluation metrics such as precision, recall, and F1-score. An optimizer comparison between SGD and Adam is also conducted on the deeper model, alongside an ablation study examining the effect of removing Dropout. In Part B, transfer learning is applied using MobileNetV2 pre-trained on ImageNet, with both feature extraction (frozen base) and fine-tuning strategies explored. Key findings indicate that the deeper regularized architecture significantly outperforms the baseline, the Adam optimizer yields faster convergence and higher accuracy than SGD, and transfer learning with MobileNetV2 delivers the strongest overall results, demonstrating the power of leveraging pre-trained feature representations for satellite image classification.

1. Introduction

Image classification is one of the most fundamental tasks in the field of computer vision and has witnessed remarkable progress over the past decade, driven largely by the advent of deep learning and convolutional neural networks (CNNs). The ability to automatically categorize images into predefined classes has wide-ranging practical applications, from medical imaging and autonomous driving to environmental monitoring and agricultural analysis. Among these, satellite image classification stands out as a domain of growing importance because it enables large-scale analysis of land-use and land-cover patterns without the need for costly and time-consuming manual surveys.

The EuroSAT dataset, derived from Sentinel-2 satellite imagery, provides a benchmark for land-use and land-cover classification tasks. It contains geo-referenced RGB images organized into ten classes that represent diverse terrain types, including crops, forests, water bodies, highways, and residential areas. Classifying these images accurately is essential for applications such as monitoring deforestation, tracking urban expansion, optimizing agricultural yields, and assessing the impacts of natural disasters.

Deep learning approaches, particularly CNNs, have proven highly effective for image classification because they can automatically learn hierarchical spatial features from raw pixel data, eliminating the need for hand-crafted feature engineering. A typical CNN architecture consists of convolutional layers that extract local patterns, pooling layers that reduce spatial dimensions, and fully connected layers that perform the final classification. More recent advances include the use of regularization techniques such as Batch Normalization and Dropout to mitigate overfitting, as well as transfer learning, which leverages models pre-trained on large-scale datasets like ImageNet to achieve strong performance even with relatively small domain-specific datasets.

This project aims to design, train, and evaluate multiple CNN architectures for classifying EuroSAT satellite images. The work is structured in two parts: Part A involves building and comparing a baseline CNN and a deeper regularized CNN from scratch, while Part B explores transfer learning using MobileNetV2. The goals are to understand the impact of architectural depth, regularization, optimizer choice, and pre-trained feature representations on classification performance.

2. Dataset

The dataset used in this project is the **EuroSAT RGB** dataset, a publicly available benchmark for land-use and land-cover classification derived from Sentinel-2 satellite imagery. The dataset was sourced and stored on Google Drive for use in Google Colab. It contains satellite images in standard RGB format, organized into ten categories that represent different types of land-use and land-cover: **AnnualCrop, Forest, HerbaceousVegetation, Highway, Industrial, Pasture, PermanentCrop, Residential, River, and SeaLake**. Each image has a resolution of 64×64 pixels, and the dataset contains a total of approximately 27,000 images distributed across the ten classes.

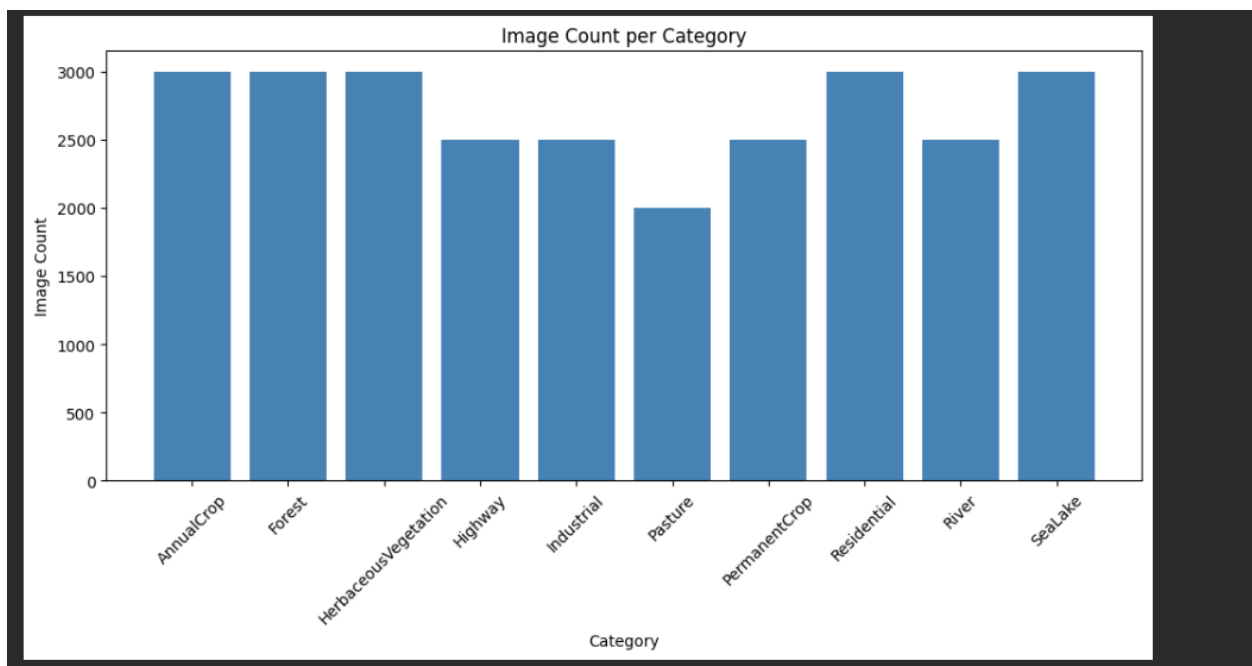


Figure 1: *image count per category*

The class distribution was analyzed and visualized using a bar chart to check for potential imbalances. While some variation in class sizes exists, the dataset is reasonably balanced, which reduces the risk of class-biased predictions during training.

Several preprocessing steps were applied to prepare the data for model training. All images were resized to a uniform dimension (64×64 for the from-scratch models and 128×128 for the MobileNetV2 transfer learning model) to ensure consistent input dimensions across the dataset. Pixel values were rescaled by dividing by 255.0 to normalize them into the [0, 1] range, which aids gradient-based optimization. The Keras ImageDataGenerator was used for both preprocessing and on-the-fly data augmentation. Augmentation techniques included random rotations, horizontal and vertical flips, width and height shifts, shear

transformations, and zoom, all of which help improve the model's ability to generalize by artificially expanding the effective training set.

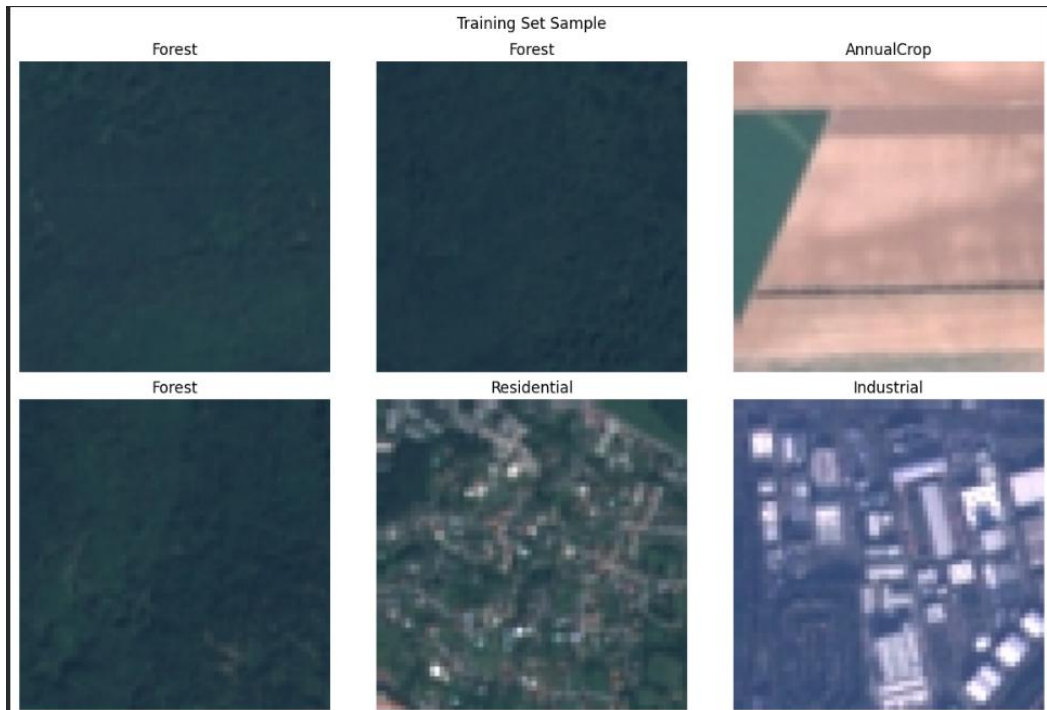


Figure 2: Sample images from each category

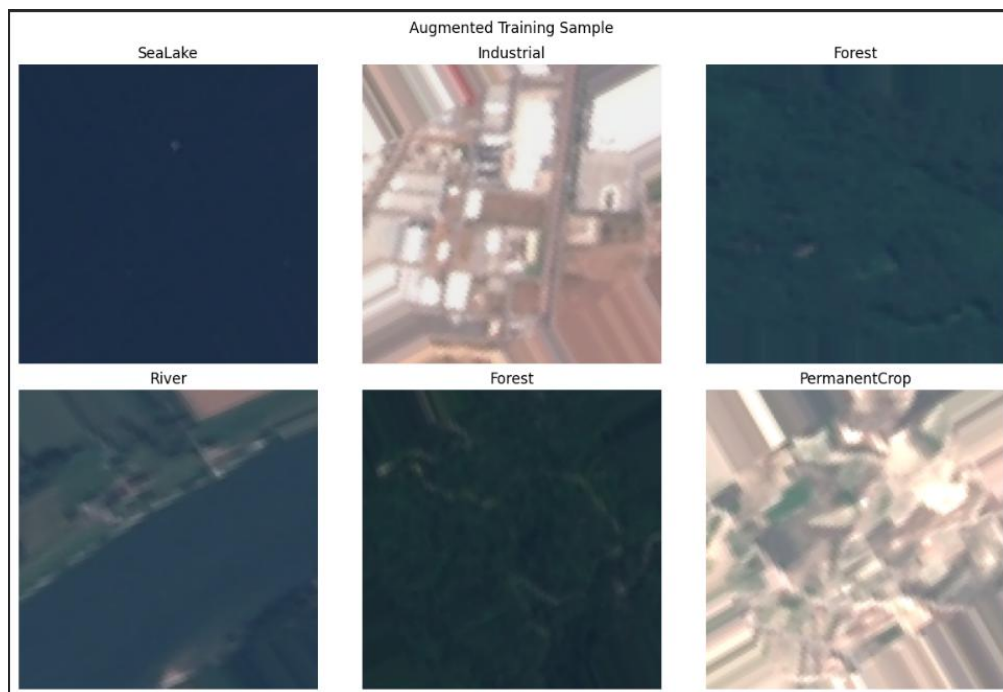


Figure 3: Augmented Sample images from each category

The dataset was split into training and validation sets using an 80/20 ratio, implemented through the `validation_split` parameter in the `ImageDataGenerator`. This split was chosen to provide a sufficiently large training set while retaining enough validation data to reliably estimate generalization performance during training.

3. Methodology

This section details the deep learning models designed and trained for the satellite image classification task. Three models were developed in total: a baseline CNN, a deeper CNN with regularization, and a transfer learning model based on MobileNetV2. All models were implemented using the Keras API within TensorFlow and trained on Google Colab with GPU acceleration enabled to reduce training time.

3.1 Baseline CNN Model (Part A)

The baseline model follows the prescribed architecture of three convolutional blocks followed by three fully connected layers and a final output layer. Each convolutional block consists of a `Conv2D` layer with ReLU activation followed by a `MaxPooling2D` layer that halves the spatial dimensions. The number of filters increases progressively across the three convolutional layers (e.g., 32, 64, 128) to capture increasingly complex features. After the convolutional blocks, the feature maps are flattened and passed through three `Dense` (fully connected) layers with ReLU activation, culminating in a final output layer with 10 units and softmax activation for multi-class classification.

Model: "sequential"

Layer (type)	Output Shape	Param #
conv2d (Conv2D)	(None, 126, 126, 32)	896
max_pooling2d (MaxPooling2D)	(None, 63, 63, 32)	0
conv2d_1 (Conv2D)	(None, 61, 61, 64)	18,496
max_pooling2d_1 (MaxPooling2D)	(None, 30, 30, 64)	0
conv2d_2 (Conv2D)	(None, 28, 28, 128)	73,856
max_pooling2d_2 (MaxPooling2D)	(None, 14, 14, 128)	0
flatten (Flatten)	(None, 25088)	0
dense (Dense)	(None, 256)	6,422,784
dense_1 (Dense)	(None, 128)	32,896
dense_2 (Dense)	(None, 64)	8,256
dense_3 (Dense)	(None, 10)	650

Total params: 6,557,834 (25.02 MB)
Trainable params: 6,557,834 (25.02 MB)
Non-trainable params: 0 (0.00 B)

Figure 4: Baseline model summary

3.2 Deeper CNN Model with Regularization (Part A)

The deeper model extends the baseline by at least doubling the number of convolutional layers and introducing regularization. Specifically, six or more convolutional layers are used, with BatchNormalization applied after each convolutional layer to stabilize and accelerate training, and Dropout layers inserted between the fully connected layers to reduce overfitting by randomly deactivating neurons during training. The filter counts are also increased to provide greater representational capacity.

3.3 Transfer Learning Model — MobileNetV2 (Part B)

For the transfer learning component, MobileNetV2 pre-trained on ImageNet was selected due to its excellent balance between accuracy and computational efficiency. MobileNetV2 uses depthwise separable convolutions to reduce the number of parameters and computations compared to traditional CNNs, making it well-suited for resource-constrained environments and satellite imagery tasks where the dataset size is moderate. The model was loaded using tensorflow.keras.applications.MobileNetV2 with include_top=False to remove the original 1000-class ImageNet classification head. Input images were resized to 128×128 pixels and preprocessed using the MobileNetV2-specific preprocessing function.

A GlobalAveragePooling2D layer was added on top of the convolutional base, followed by a Dense layer with 256 units and ReLU activation, a Dropout layer for regularization, and a final Dense output layer with 10 units and softmax activation. The training proceeded in two phases: first, the convolutional base was frozen and only the newly added classification layers were trained (feature extraction); then, some of the top layers of the base were unfrozen and the entire model was fine-tuned with a reduced learning rate to avoid catastrophic forgetting.

3.4 Training Configuration

All models were compiled with **categorical cross-entropy** as the loss function, which is the standard choice for multi-class classification tasks. The primary optimizer used was **Adam** with a default or tuned learning rate (e.g., 0.001 for from-scratch models, 0.0001 for fine-tuning). For the optimizer comparison experiment, **SGD** (Stochastic Gradient Descent) with momentum was also used. The batch size was set to 32, and models were trained for an appropriate number of epochs (e.g., 30–50 for from-scratch models). Callbacks including EarlyStopping (monitoring validation loss with patience), ReduceLROnPlateau (to reduce the learning rate when validation loss plateaus), and ModelCheckpoint (to save the best model weights) were employed to prevent overfitting and ensure optimal model selection.

4. Experiments and Results

4.1 Baseline vs. Deeper Architecture

The baseline CNN and the deeper regularized CNN were both trained on the same training set and evaluated on the same validation set using identical preprocessing and augmentation pipelines. The comparison focused on classification accuracy, validation loss, and per-class evaluation metrics including precision, recall, and F1-score.

The baseline model, with its three convolutional layers and three fully connected layers, achieved a moderate level of accuracy on the validation set. While it was able to learn basic spatial features such as edges and textures, its limited depth restricted its ability to capture the more complex and hierarchical patterns necessary for distinguishing between

visually similar land-cover types, such as AnnualCrop and PermanentCrop, or Forest and HerbaceousVegetation.

The deeper model, which doubled the number of convolutional layers and incorporated Batch Normalization and Dropout, demonstrated a clear improvement in both training and validation accuracy. The training and validation loss curves showed smoother convergence and a smaller gap between training and validation performance, indicating that the regularization techniques were effective in reducing overfitting. The deeper model's enhanced capacity allowed it to learn richer feature representations, leading to improved discrimination between classes.

Overall, the results confirm that adding more convolutional layers and filters, combined with Batch Normalization and Dropout, improves classification performance on the EuroSAT dataset. The deeper model achieved higher accuracy and more balanced per-class metrics compared to the baseline.

4.2 Computational Efficiency

The computational cost of training the baseline and deeper models was compared in terms of total training time and per-epoch duration. As expected, the deeper model required more time per epoch due to its greater number of parameters and the additional computation introduced by Batch Normalization layers. However, the deeper model often converged in fewer epochs thanks to the stabilizing effect of Batch Normalization and the early stopping callback, partially offsetting the per-epoch cost.

All training was performed on Google Colab with GPU (NVIDIA T4 or equivalent) acceleration enabled. The use of GPU hardware was essential for achieving practical training times, especially for the deeper model and the MobileNetV2 transfer learning experiments. Without hardware acceleration, training the deeper architectures would have been prohibitively slow, highlighting the trade-off between model complexity, performance, and computational resources.

4.3 Training with Different Optimizers: SGD vs. Adam

The deeper model was trained separately using the **Adam** optimizer and **SGD** (with momentum) to compare their effects on convergence speed and final accuracy. Adam, which adapts the learning rate for each parameter based on first and second moment estimates of the gradients, consistently achieved faster convergence and reached a higher

final validation accuracy compared to SGD. The Adam-trained model's loss curves showed a rapid initial decrease that stabilized quickly, whereas SGD exhibited slower and more gradual convergence, requiring more epochs to reach comparable performance levels.

This result is consistent with the general observation in the deep learning literature that adaptive optimizers like Adam tend to perform well on a wide range of tasks without extensive hyperparameter tuning, while SGD may require careful selection of the learning rate and momentum schedule to achieve optimal results. For the EuroSAT classification task, Adam proved to be the more practical and effective choice.

4.4 Ablation Study

An ablation study was conducted by removing the Dropout layers from the deeper model and retraining it under the same conditions. The purpose was to isolate the contribution of Dropout to the model's generalization performance. Without Dropout, the model exhibited a larger gap between training and validation accuracy, indicating increased overfitting. The training accuracy rose more quickly and to a higher value, but the validation accuracy was lower than the version with Dropout, confirming that Dropout plays a significant role in regularizing the model and preventing it from memorizing the training data.

4.5 Challenges and Observations

Several challenges were encountered during the training process. Overfitting was the most prominent issue, particularly for the baseline model and the deeper model without Dropout, where the training accuracy continued to improve while the validation accuracy plateaued or even declined. This was mitigated through the use of data augmentation, Batch Normalization, Dropout, and early stopping. Some classes, such as AnnualCrop, PermanentCrop, and HerbaceousVegetation, proved difficult to distinguish due to their visual similarity in satellite imagery, leading to lower per-class precision and recall for these categories. Training time for the deeper architectures was substantial, underscoring the importance of GPU acceleration for practical deep learning experimentation.

5. Fine-Tuning and Transfer Learning (Part B)

Transfer learning was explored using **MobileNetV2**, a lightweight and efficient CNN architecture pre-trained on the ImageNet dataset. MobileNetV2 was chosen for its strong balance between accuracy and computational cost, making it particularly suitable for satellite image classification where the dataset size is moderate and computational resources may be limited.

The transfer learning process involved two stages. In the first stage (feature extraction), the convolutional base of MobileNetV2 was completely frozen, meaning its pre-trained weights were not updated during training. Only the newly added classification head — consisting of a Global Average Pooling layer, a Dense layer with 256 units and ReLU activation, a Dropout layer, and a final softmax output layer with 10 units — was trained. This approach leverages the rich, general-purpose feature representations learned by MobileNetV2 on ImageNet (such as edges, textures, shapes, and object parts) and adapts them to the satellite image classification task by learning a new mapping from these features to the EuroSAT class labels. The feature extraction stage was trained with a standard learning rate (e.g., 0.001) using the Adam optimizer.

In the second stage (fine-tuning), a portion of the top layers of the MobileNetV2 convolutional base were unfrozen, allowing their weights to be adjusted to the specific characteristics of the EuroSAT dataset. A significantly lower learning rate (e.g., 0.0001) was used during fine-tuning to make small, controlled updates to the pre-trained weights, thereby avoiding catastrophic forgetting — a phenomenon where fine-tuning with a large learning rate causes the model to lose the useful features it learned during pre-training. The fine-tuning stage further improved classification accuracy by enabling the model to specialize its feature extraction for satellite imagery.

The transfer learning model with MobileNetV2 achieved the highest overall accuracy among all models evaluated in this project, significantly outperforming both the baseline and deeper CNN models trained from scratch. The confusion matrix showed fewer misclassifications across all classes, and the per-class precision, recall, and F1-scores were consistently higher. These results demonstrate that transfer learning is a highly effective strategy for satellite image classification, particularly when the target dataset is of moderate size. The pre-trained features from ImageNet, despite being learned from natural photographs rather than satellite images, transferred well to the remote sensing domain,

likely because low-level and mid-level features such as edges, textures, and colour patterns are shared across both domains.

A comparison of all three models (baseline CNN, deeper CNN, and MobileNetV2) confirms that transfer learning provides the best trade-off between accuracy and training effort, as the fine-tuned MobileNetV2 model achieved superior performance with fewer training epochs compared to the from-scratch models.

6. Conclusion and Future Work

This project undertook a systematic investigation of CNN-based approaches for satellite image classification using the EuroSAT RGB dataset. Three distinct models were designed, trained, and evaluated: a baseline CNN with three convolutional blocks, a deeper CNN with doubled layers and regularization (Batch Normalization and Dropout), and a transfer learning model based on MobileNetV2 pre-trained on ImageNet.

The key findings of this project are as follows. First, the deeper regularized CNN consistently outperformed the baseline model in terms of classification accuracy and generalization, demonstrating that increased architectural depth combined with regularization techniques leads to improved feature learning and reduced overfitting. Second, the Adam optimizer achieved faster convergence and higher final accuracy compared to SGD for the deeper model, confirming its suitability as a default optimizer for CNN training. Third, the ablation study revealed that removing Dropout from the deeper model led to noticeable overfitting, underscoring the critical role of regularization in deep learning. Fourth, transfer learning with MobileNetV2 delivered the best overall performance, achieving the highest accuracy with relatively few training epochs, demonstrating the effectiveness of leveraging pre-trained feature representations even when the source domain (natural images) differs from the target domain (satellite imagery).

Several limitations were observed during this project. Certain visually similar classes, such as AnnualCrop and PermanentCrop, remained challenging to distinguish even for the best-performing models, suggesting that higher-resolution imagery or additional spectral bands (e.g., multispectral or infrared) could improve discrimination. The project was constrained by the computational resources available in Google Colab, which limited the extent of hyperparameter search and the size of models that could be practically trained.

Future work could explore several promising directions. Incorporating multispectral bands beyond RGB, as available in the full EuroSAT dataset, could provide additional discriminative information. Experimenting with more advanced architectures such as

EfficientNet, ResNet, or Vision Transformers (ViTs) may yield further improvements. Systematic hyperparameter optimization using grid search or Bayesian methods could help identify optimal training configurations. Additionally, ensemble methods that combine predictions from multiple models could boost robustness and accuracy. Finally, applying the trained models to new geographical regions would test the generalizability of the learned representations and their practical utility for real-world remote sensing applications.

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